

Mohamed Hossam

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Portfolio: <https://mohossam.netlify.app> | GitHub: <https://github.com/mo-hossam-stack> | LinkedIn: <https://www.linkedin.com/in/mohamed-hossam-dev>

SUMMARY

Performance-driven Backend Developer specializing in building secure, production-ready web systems using Python. Expert in implementing Service Layer Architecture and database optimization to build maintainable, scalable applications. Currently preparing for GSoC 2026 to contribute to open-source ecosystems with a focus on high-concurrency backend systems.

TECHNICAL SKILLS

Backend & Architecture: Django, Django REST Framework, FastAPI, Service Layer Architecture, System Design, RESTful APIs, Asynchronous Programming.

Security & Auth: JWT/OAuth2, Session-based Auth, RBAC, CSRF/XSS Mitigation, Signed URLs (Cloudinary), Resource Hardening.

Databases: PostgreSQL, MySQL, Redis, Schema Design.

DevOps & Tools: Docker (Multi-stage builds), Nginx, Gunicorn, Uvicorn, Linux CLI, Git, Postman, CI/CD.

Languages: Python (Expert), JavaScript (ES6+), C++ (Expert), SQL.

TECHNICAL PROJECTS

CourseHub – Scalable Course Platform | *Django, PostgreSQL, HTMX, Cloudinary* | <https://github.com/mo-hossam-stack/coursehub-django> | 2025 - Present

- Designed and developed a production-ready course platform capable of serving 10,000+ students with secure, session-based access control.
- Integrated Cloudinary CDN for scalable media delivery, including private video streaming with signed URLs, adaptive bitrate streaming, and automatic image optimization (WebP/AVIF).
- Built a layered backend architecture (Views -> Service Layer -> ORM) to isolate business logic and improve maintainability.
- Implemented passwordless email authentication using UUID-based verification tokens and OTP with expiration handling.
- Designed a normalized PostgreSQL schema with indexed lookup fields to optimize concurrent access.
- Containerized the application using Docker and deployed with Gunicorn behind Nginx.

FastAPI OCR Microservice | *FastAPI, Tesseract, Docker, Pydantic v2* | <https://github.com/mo-hossam-stack/fastapi-ocr-microservice> | 2026 - Present

- Engineered a high-performance, stateless OCR microservice using non-blocking I/O for maximum throughput.
- Implemented resource hardening including a 10MB file size limit and 30-second request timeout.
- Developed structured JSON logging and health/readiness probes for production observability.
- Architected a containerized stack using Docker Compose and Nginx, ensuring horizontal scalability and environment parity.

Clinic Management System (CMS) | *Django, PostgreSQL* | https://github.com/mo-hossam-stack/orthodontic_patient_portal | 2025 - Present

- Engineered a comprehensive web-based dental practice management system for handling patients, visits, payments, and medical records in a unified workflow.
- Implemented secure Role-Based Access Control (RBAC) using Django authentication to restrict access to sensitive healthcare data.
- Designed a normalized relational PostgreSQL schema supporting patients, visits, payments, and X-ray records with optimized data relationships.

- Built end-to-end patient lifecycle management including registration, visit scheduling, treatment documentation, and payment tracking.
- Developed automated outstanding balance calculations and real-time payment status updates to improve financial transparency.
- Integrated secure medical image handling with organized X-ray uploads, visit linking, and protected media storage.

Chess Game | *Python, Pygame* | <https://github.com/mo-hossam-stack/python-pygame-chess> | 2025 - Present

- Developed a modular chess game with full move validation, check/checkmate detection, and pawn promotion logic.
- Implemented turn-based gameplay with a graphical board interface and complete game state handling.

EDUCATION

B.Sc. in Computer Science | Nahda University, Beni Suef | GPA: 3.23 / 4.0 | Expected Graduation: 2028

CERTIFICATIONS

- ALX Professional Foundations
- ALX Ventures Freelance Academy
- ICPC ECPC Qualifications – Honorable Mention

ACHIEVEMENTS

- Ranked 20th in Constructor Open Cup 2026 – annual international programming competition organized by Constructor University & JetBrains Foundation.
- Solved 1,100+ algorithmic problems across Codeforces, LeetCode, and VJudge.
- Achieved Expert rating on Codeforces.

ADDITIONAL INFORMATION

Languages: Arabic (Native), English (Fluent – B2).